Food Remedy API

Cross Referencing E-Numbers with Allergen Categories – RE033

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Introduction

In the European Union, the European Food Safety Authority (EFSA) assesses the safety of food additives, assigning each an E-number once it has successfully undergone rigorous testing and demonstrated a valid technological function without misleading consumers. Labels are required to show both the purpose of the additive, such as color or preservative, and its specific identity, either by name or E-number. Additives are substances not typically consumed alone but intentionally added for technological reasons, with examples ranging from synthetic chemicals to natural compounds like vitamin C (E300) and lecithin (E322). A similar system is used in Australia and New Zealand under Food Standards Australia New Zealand (FSANZ), where additives undergo hazard and dietary exposure assessments to determine the Acceptable Daily Intake (ADI). This ADI is calculated by applying a safety factor of 100 to the highest level at which no adverse effects are observed in animal studies, ensuring human consumption stays within safe limits. These frameworks together demonstrate a shared international commitment to protecting public health while allowing the responsible use of food additives.

Standard Classification by Functional Category

The E-number system offers a structured classification based on the additive's primary technological function. This high-level grouping is the first and most basic layer of the proposed taxonomy. The established numeric ranges for these categories are a useful starting point for data organization:

- **E100-E199 (Colours):** Substances added to enhance or restore colour. This category includes well-known additives like Tartrazine (E102), Sunset Yellow FCF (E110), and the animal-derived Cochineal (E120).
- **E200-E299 (Preservatives):** Chemicals that prevent or slow the growth of microbes, thereby extending shelf life. Examples include Sulphur Dioxide (E220) and Sodium Benzoate (E211).
- **E300-E399 (Antioxidants and Acidity Regulators):** These additives prevent oxidation and help maintain the proper acid level in foods. Lecithins (E322) and Vitamin C (E300) are part of this group.
- **E400-E499 (Thickeners, Stabilisers, Emulsifiers):** This broad category includes substances that alter a food's texture and consistency. Examples include Carrageenan (E407) and Guar Gum (E412).

- **E600-E699 (Flavour Enhancers):** Additives that increase the power of a food's existing flavour, such as monosodium glutamate (MSG).

Regulatory approval for food additives is not permanent but subject to ongoing review. A clear example of this is Titanium Dioxide (E171), which was initially permitted for use as a food additive. However, following a re-assessment by the European Food Safety Authority (EFSA), it was concluded that the substance could no longer be regarded as safe. Consequently, its approval was withdrawn, and it was banned across the European Union. This case illustrates the evolving nature of food safety regulation, where scientific evidence continually informs policy decisions aimed at protecting public health.

Allergen and Health-Risk Taxonomy

Drawing upon both regulatory and scientific principles, the following section presents a detailed taxonomy of high-priority food additives. This framework progresses from broad functional groupings towards a more specific, health-focused classification, allowing for greater clarity and precision in understanding their impact.

The Sulphite Group (E220-E228)

- **Functional Group:** Preservatives
- **Chemical Group/Family:** Sulphites
- **Associated Health Risks:** These additives pose a high risk of respiratory allergens, particularly for individuals with asthma. The mechanism of the reaction is typically a sensitivity, not a true allergy, as it does not involve the immune system. It is theorized to be caused by the inhalation of sulfur dioxide gas released by the sulfites upon ingestion, which can cause airways to constrict.
- **Known Symptoms:** include wheezing, shortness of breath, coughing, chest tightness, hives, and, in rare cases, anaphylaxis.

The Benzoate Group (E210-E219)

- **Functional Group:** Preservatives.
- **Chemical Group/Family:** Benzoates
- **Associated Risks:** Hypersensitivity. The research also points to a potential link between sodium benzoate (E211) and hyperactivity in children, a connection noted in the "Southampton study" alongside artificial colours.
- **Known Symptoms:** Skin irritation, hives, perioral contact urticaria, and potential exacerbation of asthma symptoms.

Artificial Colours: The "Southampton Six" & Others

- **Functional Group:** Colours.
- **Chemical Group/Family:** Varies (e.g., Azo Dyes).

- **Associated Risks:** A significant body of research suggests a link between certain artificial colours and hyperactivity and neurobehavioral issues in children. The "Southampton Six" (E102, E104, E110, E122, E124, E129) are the primary members of this group. Additionally, some colours, such as Annatto (E160b) and Carmine (E120), have documented links to true allergic reactions, including anaphylaxis. Tartrazine (E102) itself is linked to hives, asthma symptoms, and is noted for its cross-reactivity with other substances.
- **Known Symptoms:** Hyperactivity, restlessness, irritability, skin rashes, hives, and asthma.

Other Noted Sensitivities and Consumer Concerns

The research also identifies several other additives with notable health or dietary implications:

- **Guar Gum (E412):** A thickener that can trigger rare allergic reactions or rhinitis, and has been linked to occupational asthma.
- **Carrageenan (E407):** A gelling agent linked to adverse gastrointestinal effects in some individuals.
- **Lecithin (E322):** An emulsifier that, while generally safe, is often derived from common allergens such as soy and eggs, necessitating caution for individuals with these specific allergies.
- **Animal-Derived Additives:** Beyond allergens, the taxonomy must also account for dietary concerns. Additives such as Cochineal (E120), Gelatine (E441), and Edible Bone Phosphate (E542) are always of animal origin. Public datasets, such as the E-Number database on GitHub, provide valuable information on the Halal status of additives, which is another vital tagging requirement for a comprehensive system.

E-Number Functional Classification Table

E-Number Range	Functional Category	Description/Examples
E100–E199	Colours	Additives that add or restore color to foods. Includes Tartrazine (E102) and Allura Red AC (E129).
E200–E299	Preservatives	Substances that prevent or slow the deterioration of food by micro-organisms. Includes Sulphur Dioxide (E220) and Sodium Benzoate (E211).
E300–E399	Antioxidants & Acidity Regulators	Prevent oxidative deterioration and maintain proper acid levels. Includes Vitamin C (E300) and Lecithins (E322).
E400–E499	Thickeners, Stabilisers, Emulsifiers	Additives that increase viscosity, maintain uniform dispersion, and facilitate the mixing of oil and water. Includes Guar Gum (E412) and Carrageenan (E407).
E600–E699	Flavour Enhancers	Substances that enhance a food's existing taste and odor. Includes Monosodium Glutamate (MSG).

High Priority Additive Profiles and Taxonomy

E-Number	Chemical Name	Functional Group	Chemical Group/Family	Associated Health Risks	Known Symptoms	Dietary Concerns
E102	Tartrazine	Colour	Azo Dye	Hypersensitivity, hyperactivity, cross-reactivity with aspirin & salicylates	Hives, skin rashes, asthma, hyperactivity	N/A
E120	Cochineal; Carminic acid	Colour	Anthraquinone	True allergy, anaphylaxis	Hives, swelling, anaphylaxis	Non-vegan, non-vegetarian
E211	Sodium benzoate	Preservative	Benzoate Group	Hypersensitivity, potential link to hyperactivity	Hives, skin irritation, asthma symptoms	N/A
E220	Sulphur dioxide	Preservative	Sulphite Group	High-risk respiratory allergen for asthmatics	Wheezing, shortness of breath, hives, rare anaphylaxis	N/A
E322	Lecithins	Emulsifier	Phospholipid	Potential for allergenic proteins from source (soy, egg)	Allergic reactions related to source	Depends on source material
E412	Guar gum	Thickener	Natural Gum	Rare allergy, rhinitis, occupational asthma	Hives, rhinitis, gastrointestinal issues	N/A

Detailed E-Number Profiles

E102 (Tartrazine)

E102, or Tartrazine, is a petroleum-derived azo dye widely used to colour foods such as drinks, confectionery, mustard, and baked goods. Although generally approved as safe, it has been associated with adverse effects in sensitive individuals, including skin reactions, aggravation of neurodermatitis in children, and asthma-like symptoms in susceptible groups. Tartrazine is also part of the “Southampton Six,” additives linked to hyperactivity and attention deficits in children, which has led the EU to require warning labels on products containing it. Scientific studies suggest that these effects are more pharmacological than immunological, with familial susceptibility and cross-reactivity with other azo dyes and natural compounds such as salicylates frequently observed.

E211 (Sodium Benzoate)
Sodium Benzoate (E211) is a common preservative used in foods such as jams, pickles, and salad dressings. While it is generally regarded as safe, a small proportion of people may experience hypersensitivity, which can lead to allergic reactions such as hives, skin irritation, or asthma-like symptoms, with rare cases progressing to anaphylaxis. Research has also raised concerns about a possible association between Sodium Benzoate and hyperactivity in children, though further studies are required to establish this link. Overall, while harmless for most, its effects on sensitive individuals highlight the need for continued monitoring and scientific evaluation.

E220 (Sulphur Dioxide)
Sulphur Dioxide (E220) is a widely used preservative that helps prevent foods such as dried fruits, wines, and potatoes from browning. Although safe for most people, it is recognised as a high-risk substance for individuals with asthma, as it can provoke respiratory reactions including wheezing, chest tightness, and shortness of breath. These effects are caused by sensitivity rather than true allergy, resulting from inhaling the gas released when the additive is consumed. While most cases are mild, severe reactions, including anaphylaxis, have occasionally been reported. To reduce risks, regulations require that products containing sulfites be clearly labelled, allowing sensitive individuals to avoid them effectively.

Conclusion

This report proposes a comprehensive taxonomy for food additives that integrates regulatory data, scientific findings, and consumer health risks into a structured model for backend tagging. The framework extends beyond simple E-number listings, offering a multi-layered approach that enhances accuracy and usability. It recommends the immediate implementation of a core database focusing on high-priority additives such as Sulphites, Benzoates, and Artificial Colours, which are strongly associated with adverse health effects. To ensure reliability, the system must include continuous monitoring mechanisms, allowing updates in response to regulatory changes such as the ban of Titanium Dioxide (E171). Future development should incorporate advanced features linking additives by chemical groups or cross-reactive substances, thereby providing users with more intelligent safety alerts. Finally, the report highlights the importance of sourcing data primarily from official authorities like EFSA and FSANZ, while treating public datasets as supplementary references. This strategy ensures that the system remains scientifically credible, adaptable, and aligned with international food safety standards.